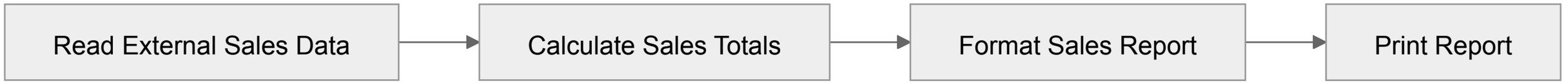
Why Use Databases?

Modern applications and the state they need to remember

Gregory S. DeLozier, PhD

Working with External Data

A sales export is input to an analysis.



Each run ends with the report.

Persisting Application State

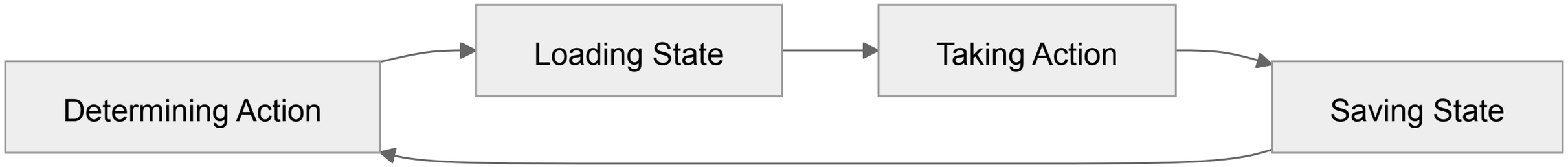
Objects in memory disappear when the program ends.

The order still needs to exist tomorrow.

What Needs to Survive?

Keep after a restart	Usually temporary
An order’s contents	A value used to draw a button
A pet’s owner	An intermediate calculation

Persistence Cycle



Display State can happen anywhere in this cycle.

A read-only action may load and display without saving.

Why Not Just a File?

A file may be enough for a small program.

- Several people change the same information.
- A change fails partway through.
- Finding the needed records gets harder as data grows.

Pets and Owners

How does the application remember who owns a pet?

What happens when an owner is removed?

How does a web-page change become persistent?

Letting the Database Do More

The page needs the number of dogs.

Application counts	Database counts
Retrieve every pet	Request the count
Count dogs in the application	Receive the result

Course Outline: SQL and the Application

Structured Query Language (SQL)

Chapter	Topic
1	Intro SQL
2	SQL in Python
3	Intro to Web Apps
4	Database Abstraction
5	Database Constraints

Course Outline: Data Access and Optimization

Object-relational mapper (ORM)

Chapter	Topic
6	Object-Relational Mappers (ORMs)
7	ORM with Constraints
8	Dataset Library
9	Optimization

Course Outline: Other Platforms

Chapter	Topic
10	Alternative Platforms
11	Intro to Mongo
12	Mongo App
13	MongoDB Atlas
14	Mongo Indexes

Course Outline: Servers and Applications

Chapter	Topic
15	Moving to a SQL Server
16	Moving to a Mongo Server
17	MapReduce and Aggregation Pipelines
18	Geospatial Queries
19	MongoDB Robustness Strategies

An Application You Use

What should it remember after you close it?

What can it forget?

What work could its database do?